EAST WEST UNIVERSITY

Course Details

Course Title: Computer Networks

Couse Code: CSE 405

Section: 05

Submitted To

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Star Topology:

Star Topology is a network setup in which each device is connected to a central node called a hub (switch). The hub manages the data flow between the devices. In this topology, all the nodes in a star topology are connected to the Hub and the Hub is responsible for the transmission of the data in the network. Whenever any node wants to transmit data to another node, it first transmits data to the central node (hub), and then it transfers the data to all the nodes on the network. After receiving the data, the hub checks for the destination address, and if the address matches the data is accepted otherwise data is rejected. The figure below shows the star topology.

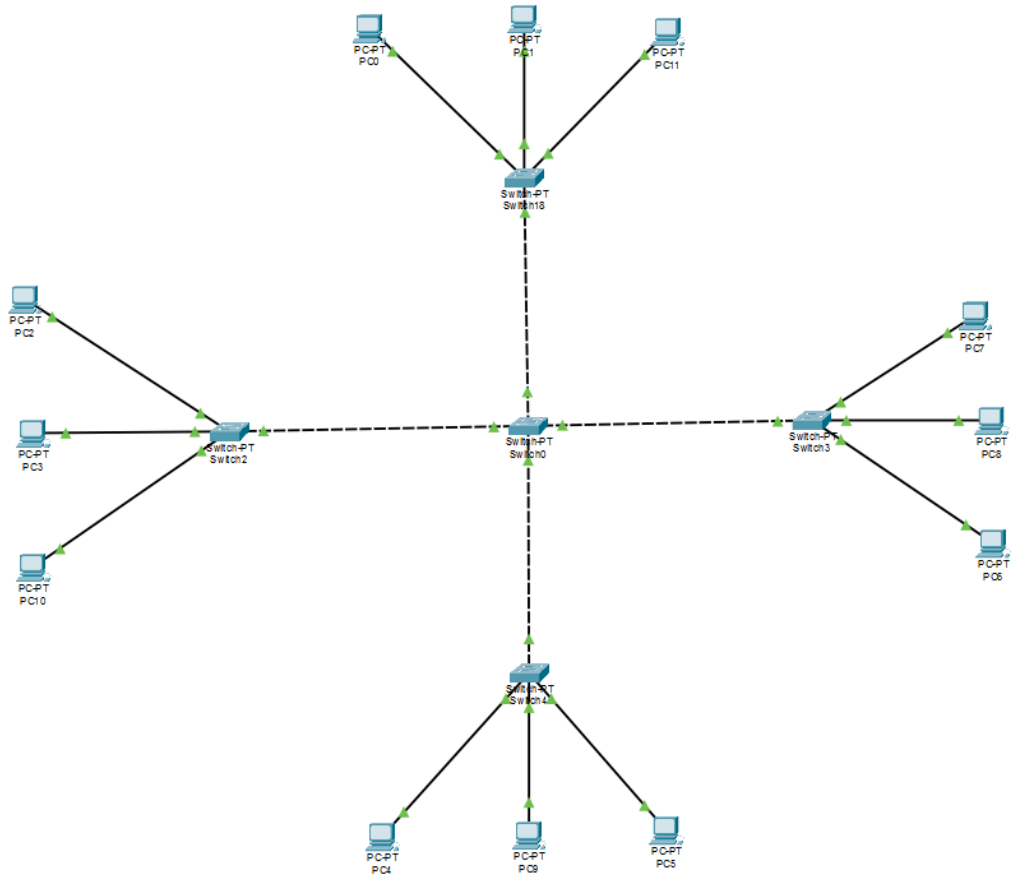

Figure 1: Star Topology

Bus Topology:

Bus topology is a network setup where each computer and network device is connected to a single cable or backbone. It requires less cable length than other topologies, such as star. In bus topology, it can be difficult to identify the problems if the whole network goes down. If a main cable is damaged, the network fails. The figure below shows the bus topology.

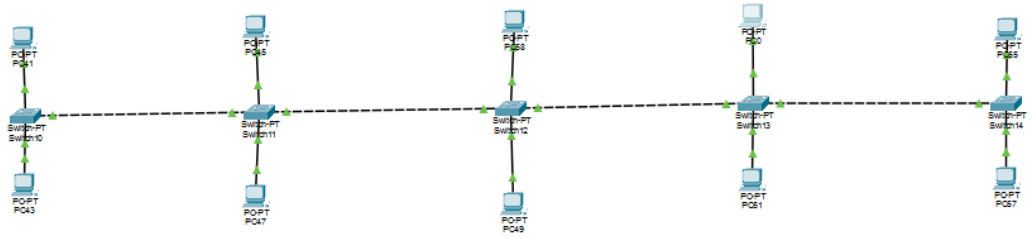

Figure 2: Bus Topology

Hybrid Topology:

A Hybrid Topology is any network that is created by connecting two or more different fundamental topologies (like Star, Bus, Ring) to form a larger, more robust network. A Star Bus Hybrid Topology is a combination of Star and Bus topologies.

Features of Star Bus Hybrid Topology

1. Local Connectivity

In this topology, each group of computers is connected to a local switch, forming a Star topology. This makes network management easier, provides isolation in case of device failure, and allows fast data transfer within that local group.

2. Backbone Connectivity

The main switches from each star network are connected through a Bus like backbone. This setup is simple and cost effective for linking multiple star networks across a large area.

3. Scalability

The Star Bus hybrid is highly scalable. New star networks can be added easily by connecting a new switch to the main bus line without disturbing the existing structure.

4.Reliability

This topology provides moderate reliability. If one local switch fails, only that star network is affected, while the rest of the system continues working.
However, if the main bus or a central switch fails, it can disrupt a large part of the network.

5. Common Use

This type of network is commonly used in large organizations or campus networks, where different departments (each having their own star network) are connected through a main bus backbone.

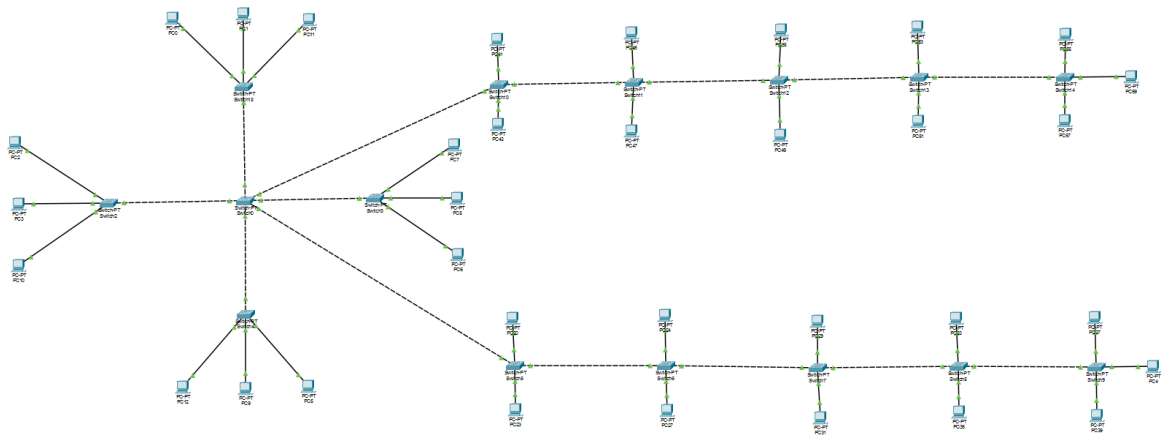

Figure 3: Hybrid Topology using Star & Bus topology.